Canberra, Australia

Hao Lu

Robotics Researcher | Planning, Manipulation & Robotics Systems

PhD Candidate in Robotics · 10+ Years of Industrial R&D Experience

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Profile

Robotics researcher and systems engineer developing reliable planning and manipulation through algorithms, optimization, learning, and integrated robotic systems. Combines research spanning theoretical foundations, algorithm design, simulation, and real-robot experimentation with over 10 years of industrial R&D at a leading semiconductor company, progressing to Senior Principal Software System Engineer. Brings expertise across computer architecture, operating systems, heterogeneous computing, and hardware–software integration, with a focus on how algorithmic and system-level decisions jointly determine performance and reliability. Research published at leading international venues in robotics.

Research focus: Reliable motion and task planning, manipulation and rearrangement, decision-making under uncertainty, learning and generative models for planning, and efficient robotic systems.

Research Experience and Selected Contributions

PhD Researcher, Australian National University Feb 2024 – Present

Develop reliable and computationally efficient robotic planning and manipulation by connecting mathematical foundations, learning and systems design. Investigate how algorithms, software architecture and heterogeneous computing jointly shape planning performance and dependable execution under uncertainty.

1. RELIABLE AND SCALABLE MOTION PLANNING

Collision-probability-aware motion planning Feb – Oct 2024

Published: IROS 2024

- Formalised a theoretical framework for motion planning under environment uncertainty, modelling whole-path collision probability in high-dimensional spaces for arbitrary uncertainty distributions accessible through sampling.
- Connected the formulation to minimum constraint removal. Established asymptotic optimality in the sampling-based regime and developed a greedy-seeded MILP for optimal planning over the sampled roadmap.

JUMP: Joint Uncertainty-aware Manipulation Planning Started Oct 2025

Ready for submission. Target: IEEE Robotics and Automation Letters (RA-L)

- Formulated joint trajectory-collision and terminal-grasp failure under shared scene uncertainty, preserving their dependence without parametric uncertainty models or independence assumptions.
- Developed a sampling-based framework and MILP to jointly select approach trajectories and grasps that minimise empirical task-failure probability, with exact optimisation over the sampled planning representation.
- Established asymptotic optimality under roadmap densification, increasing joint-world samples and appropriate grasp coverage. Integrated GPU-accelerated collision checking and physics-based grasp evaluation.

Enabling Halton-Based Extremely Dense Roadmaps Oct 2026 – Present

Ongoing

- Investigating scalable construction and representation of reusable Halton-based roadmaps, targeting millions of configurations for high-dimensional, multi-query motion planning.
- Studying when dense connectivity and reuse improve query-time reliability in narrow passages and constrained robot motions, accounting for preprocessing, memory and collision-checking costs.

2. CONTACT-RICH MANIPULATION AND REARRANGEMENT

Optimal knock–pick planning Dec 2025 – May 2026

Accepted: WAFR 2026

- Formulated a graph-based abstraction for tightly packed, grid-aligned block manipulation, combining directional knocks that create grasp clearance with parallel-gripper picks.
- Reduced minimum-knock planning to optimal exact cover via maximum-weight perfect matching, yielding a polynomial-time solver and a construction of feasible knock–pick sequences under the model assumptions.

Planning with toppling actions Mar – Oct 2025

Accepted: IROS 2026

- Introduced the SMASH capacity-constrained graph abstraction for aggregating actions that move multiple objects simultaneously. Formulated candidate task planning as a variant of pebble motion.
- Developed a mode-graph and partially grounded task-and-motion framework interleaving topple and pick–place actions, grounding uncertain outcomes during execution. Demonstrated the abstraction's applicability to scoop actions.

Contact-rich slide–pick manipulation 2026

Accepted: CoRL 2026 Workshop on Open Problems in Contact-Rich Loco-Manipulation

- Formulated minimum-slide planning for dense cuboid clutter by discretising continuous grasps into finite witnesses preserving every blocker set. Encoded dependencies as a monotone Horn system with order-independent pick closure and a minimum-key characterisation.
- Proposed an exact MILP for geometric instances and validated the integrated perception–planning–execution pipeline across thousands of physics-simulated executions, with all real-robot trials completed successfully.

3. LEARNING-INFORMED PLANNING AND TASK REPRESENTATION

Learning Task and Motion Constraints from Demonstrations Oct 2026 – Present

Ongoing

- Developing a mathematical framework to infer structured task specifications from observations and demonstrations, connecting inferred task modes and transition conditions with shared continuous manipulation constraints.
- Investigating identifiability, applicability and composition conditions for demonstration reuse through direct replay, transformed trajectories and alternative motions satisfying the inferred task and motion constraints.

Generative world priors for partially observed rooms Aug 2025 – Apr 2026

Accepted: RSS 2026 Workshop on Robot Planning in the Era of Foundation Models

- Developed a zero-shot pipeline combining generative outpainting, depth and segmentation to convert partial RGB observations into semantically labelled 3D world samples, without task-specific fine-tuning.
- Formulated a minimum-constraint-removal objective coupling path collision and target-reaching failure across sampled worlds. Improved safety and reaching success over uniform and single-world priors on Matterport3D tasks.

Kinodynamic planning under uncertainty Aug 2026 – Present

Ongoing

- Investigate diffusion and flow-matching models of environment dynamics for forward propagation of a planner's search tree, including control/duration sampling under environment and control uncertainty.
- Study the interaction between learned dynamics and planning reliability. Asymptotic properties remain under investigation.

4. EARLIER RESEARCH AND COURSEWORK

Arbitor: numerical emulation for DNN accelerators Sep 2022 – Jan 2023

University of Toronto. Published: USENIX ATC 2023

- Co-developed an accelerator emulation tool using machine-learning compiler infrastructure, connecting hardware optimisation choices to numerical behaviour and DNN accuracy through accurate software emulation.
- The study evaluates low-precision arithmetic, approximate computing and structured sparsity using GPU-based emulation, supporting rapid accelerator evaluation before hardware is built.

Earlier projects in machine learning and computer vision 2021 – 2023

Completed coursework projects

- **Sample-efficient control:** Combined imitation learning with PILCO-style model-based policy learning.
- **Multi-object tracking:** Compared camera–LiDAR fusion with LiDAR-only tracking baselines.
- **Text classification:** Studied unsupervised classification using dictionary learning.
- **Blind source separation:** Investigated FastICA and dictionary-learning methods.
- **Dimensionality reduction:** Explored exponential-family principal component analysis.
- **Photometric stereo:** Investigated surface reconstruction under near-light illumination.
- **Motion prediction:** Investigated multimodal trajectory prediction for autonomous driving.

Industrial R&D Experience

Freescale Semiconductor / NXP Semiconductors Jul 2013 – Feb 2025

Computing Systems, Edge AI & Performance Optimization

Freescale Semiconductor Jul 2013 – Dec 2015
NXP Semiconductors, Edge Processing Business Line Dec 2015 – Feb 2025

CAREER PROGRESSION

Senior Principal Software System Engineer Sep 2021 – Feb 2025
Principal Software System Engineer Apr 2018 – Aug 2021
Senior Application Engineer Sep 2015 – Mar 2018
Application Engineer Jul 2013 – Aug 2015

SYSTEMS ENGINEERING CONTRIBUTIONS

- Diagnosed hardware/software bottlenecks using Linux perf, ftrace/trace-cmd, Arm Streamline, CodeWarrior and OProfile. Optimised algorithms, compiler output and instruction paths with attention to cache, TLB, memory/bus behaviour and multicore execution.
- Developed and integrated software across application code, operating-system interfaces, runtimes and firmware. Analysed processor and accelerator behaviour to improve system performance and resolve integration issues.
- Delivered proof-of-concept systems, application notes and technical documentation. Trained internal teams, partners and customers, and provided remote and on-site support for integration, diagnosis and deployment.

Automotive Compute Platform: Initialization & Performance Engineering

- Optimised the compute-platform startup path supporting an autonomous-car project, addressing image loading and costly MMC, PHY, Ethernet, PCIe and memory-controller initialisation across hardware/software dependencies.
- Implemented U-Boot Falcon Mode to load and start the Linux kernel directly from the Secondary Program Loader (SPL), bypassing the full U-Boot stage and addressing dependencies between kernel loading and hardware initialisation.

Edge-AI model deployment and heterogeneous computing

- **Machine Learning Foundation Project:** Collaborated with NXP's CTO organisation on internal infrastructure for creating, adapting, deploying, optimising and evaluating deep-learning/foundation models for edge AI.
- **NPU and Heterogeneous Inference:** Developed Neutron inference models and deployment workflows. Investigated execution on NPU, GPU and Arm Cortex-A and Cortex-M targets and optimisation of supporting drivers, runtimes and firmware.
- **i.MX9 Edge-AI Applications:** Developed and deployed AI algorithms on i.MX9 NPU platforms for appliance anomaly detection, diagnosis/prediction and non-intrusive load disaggregation.

Platform experience: NXP Layerscape networking processors (LS1043A, LS1046A, LS1088A, LS2088A, LX2160A) and i.MX applications processors (i.MX8 and i.MX9 series), including NPU-based inference. Experience spans networking, automotive computing applications and edge AI.

Selected Peer-Reviewed Publications

Contact-Rich Slide–Pick Manipulation Planning for Dense Cuboid Clutter with Parallel Grippers.
H. Lu, R. Shome. *CoRL 2026 Workshop on Open Problems in Contact-Rich Loco-Manipulation. Accepted.* PDF

Virtues of Ordered Chaos: Planning with Topple Actions in Tabletop Stack Rearrangement.
H. Lu, R. Shome. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. Accepted.* PDF

Optimal Knock-Pick Planning for Tightly Packed Tabletop Blocks With Parallel Grippers.
H. Lu, R. Shome. *World Symposium on the Algorithmic Foundations of Robotics (WAFR), 2026. Accepted.* PDF

Zero-Shot Generative World Priors for Planning and Object Reaching in Partially Observed Rooms.
H. Lu, S. S. Bhattacharjee, D. Campbell, R. Shome. *RSS 2026 Workshop on Robot Planning in the Era of Foundation Models (FM4RoboPlan). Accepted.* PDF

Sampling-based Motion Planning for Optimal Probability of Collision under Environment Uncertainty.
H. Lu, H. Kurniawati, R. Shome. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024, pp. 3138–3145.* DOI

Arbitor: A Numerically Accurate Hardware Emulation Tool for DNN Accelerators.
C. Jiang, A. Jayarajan, **H. Lu**, G. Pekhimenko. *2023 USENIX Annual Technical Conference (USENIX ATC '23), pp. 519–536, 2023.* Paper

Education

Australian National University (ANU), Canberra, Australia — Feb 2024 – Present
PhD Candidate in Computer Science (Robotics)
Supervisors: Dr Rahul Shome, Prof Hanna Kurniawati, Dr Marco Cheung

Australian National University (ANU), Canberra, Australia — Feb 2021 – Dec 2023
Master of Machine Learning and Computer Vision | GPA: 6.92/7.0
Recipient of the ANU Postgraduate Medal for Academic Excellence

University of Toronto, Toronto, Canada — Sep 2022 – Apr 2023
Graduate Exchange Student, Department of Computer Science | GPA: 4.0/4.0

Nankai University, Tianjin, China — Sep 2010 – Jul 2013
Master of Engineering in Computer Science / Computer Architecture

Technical Expertise

- **Planning and optimisation:** Motion/task planning, uncertainty modelling, combinatorial and graph algorithms, MILP and MIQCP, Gurobi, sampling-based methods and Monte Carlo evaluation.
- **Robotics software and simulation:** ROS2, MoveIt2, OMPL, Isaac Sim, Gazebo, collision checking and integration of planning software with simulated and physical robotic systems.
- **Operating systems and systems software:** Linux, U-Boot/SPL, boot optimisation, OS interfaces, drivers/runtimes, cross-compilation, GCC/LLVM, CMake/Make, Git, Linux perf, ftrace/trace-cmd, Arm Streamline, CodeWarrior and OProfile.
- **Computer architecture and heterogeneous computing:** Arm and PowerPC, cache/TLB, memory/bus analysis, multicore execution, SIMD/NEON and NPU inference, NXP DPAA/DPAA2 data-path architectures, AIOP programmable packet processing and NETC networking hardware.
- **Machine learning and inference:** PyTorch, TensorFlow, probabilistic and generative models, model deployment and inference optimisation, OpenCV, Boost and Eigen.
- **Programming:** C++, Python, C, Java, MATLAB, and architecture- and accelerator-specific assembly.

Selected Awards, Teaching and Academic Service

Awards and recognition

ANU PhD Scholarship and ANU HDR Fee Remission Merit Scholarship	2024
ANU Postgraduate Medal — 19 recipients among ∼7,000 graduates	2023
NXP China R&D Innovation Award	2023
NXP Top Achiever	2015 – 2022
NXP Bravo Award	2021
NXP Bravo Award	2018
Freescale Rockstar Award	2015

Teaching, Australian National University

Tutor

Algorithms (COMP3600)	Jul 2026 – Present
Advanced Topics in AI (COMP8620)	Jul 2026 – Present
Statistical Machine Learning (COMP8600)	Feb – Jun 2026
Advanced Algorithms (COMP8460)	Feb – Jun 2026
Advanced Topics in AI (COMP8620)	Jul – Nov 2025
Introduction to Machine Learning (COMP6670)	Jul – Nov 2025

Academic service

- **Journal reviewer:** IEEE Transactions on Robotics (T-RO).
- **Conference reviewer:** IROS 2025 and 2026, ICRA 2026 and 2027, Humanoids 2026, and CoRL 2026.
- **Organisation:** Helped organise AusRos 2026, Canberra.
- **Guest lecture:** Advanced Algorithms (COMP8460), ANU, Canberra, June 2026.